Important questions for DWDM MSE1

1.OLAP Operation [Unit 1]

2.Characteristics of Data Warehouse [Unit 1]

3.Difference between OLAP and OLTP [Unit 1]

4.ODS (Operational Data Store) [Unit 1]

5.Snowflake and Star Schema [Unit 1]

6.Data Mining [Unit 2]

7.Task and Motivating Challenges of Data Mining [Unit 2]

8.KDD (Knowledge Discovery in Databases) [Unit 2]

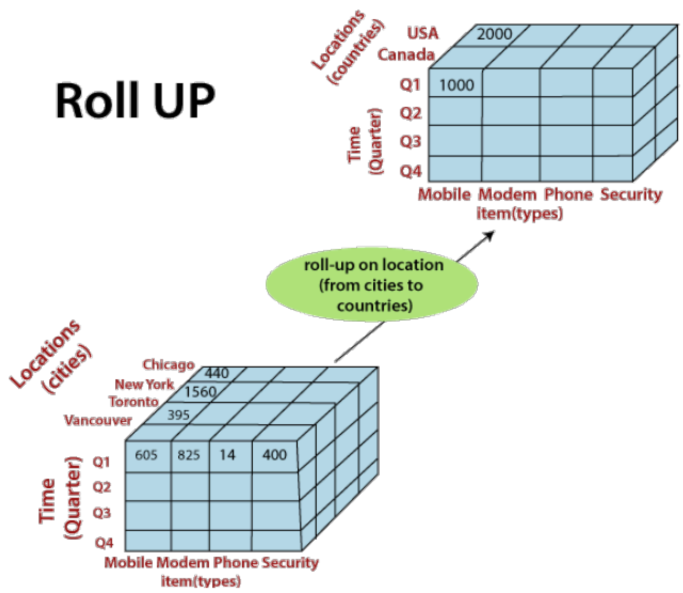
9.Types of Attributes [Unit 2]

1.OLAP Operation

•Roll-up
•Drill-down
•Slice and dice
•Pivot (rotate)

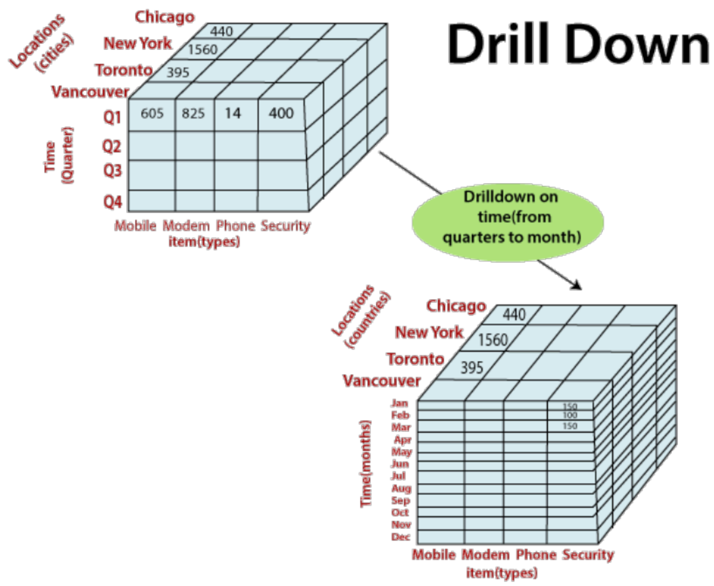
Roll-up:

→Also called drill-up operation
→Performs aggregation on a data cube, either by climbing up a concept hierarchy or by dimension reduction
→It navigates from more detailed data to less detailed data.
→Initially the concept hierarchy was "street < city < state< country".
→ On rolling up, the data is aggregated by ascending the location hierarchy from the level of city to the level of country.



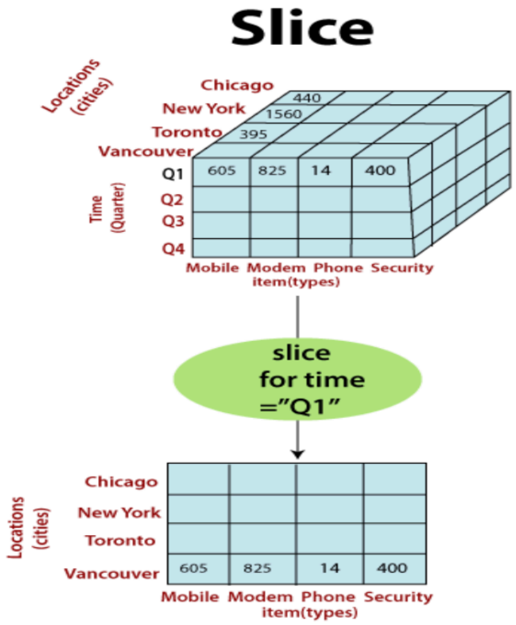
Drill-down:

→Drill-down is the reverse of roll-up.
→It navigates from less detailed data to more detailed data.
→Drill-down can be realized by either stepping down a concept hierarchy
→ Initially the concept hierarchy was "day < month < quarter < year."
→ On drilling down, the time dimension is descended from the level of quarter to the level of month.
→When drill-down is performed, one or more dimensions from the data cube are added.



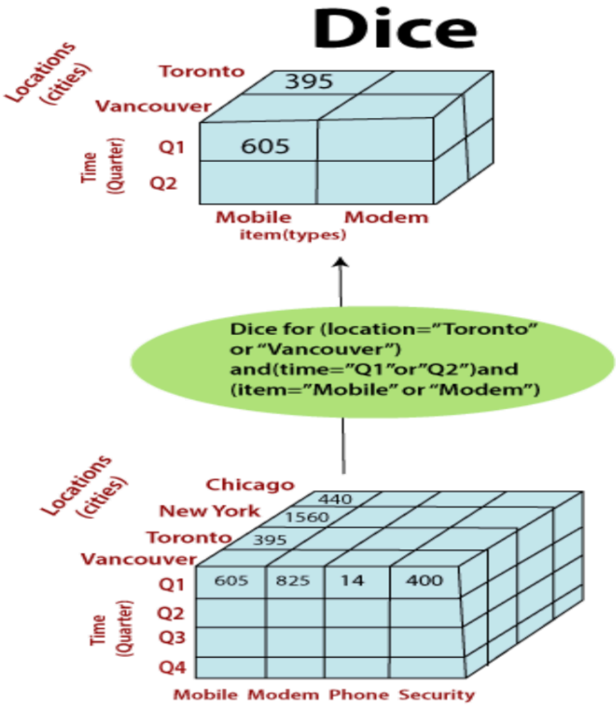
Slice and dice:

- The slice operation selects one particular dimension from a given cube and provides a new sub-cube.
- Consider the following diagram that shows how slice works.
- Here Slice is performed for the dimension "time" using the criterion time = "Q1".
- It will form a new sub-cube by selecting one or more dimensions.



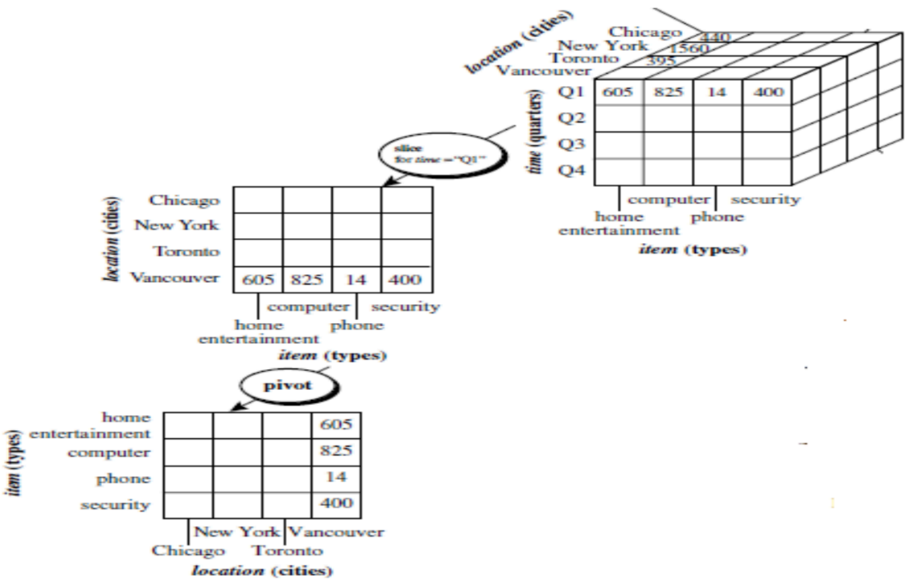
Dice selects two or more dimensions from a given cube and provides a new sub-cube.

Consider the following diagram that shows the dice operation. The dice operation on the cube based on the following selection criteria involves three dimension(location = "Toronto" or "Vancouver") (time = "Q1" or "Q2") (item =" Mobile" or "Modem")



Pivot (rotate):

- Visualization operation that rotates the data axes in view to provide an alternative data presentation



2.Characteristics of Data Warehouse

> "A data warehouse is a **subject-oriented, integrated, time-variant, and non-volatile** collection of data in support of management's decision-making process."

The four main characteristics are:

---

1) Subject-Oriented

A data warehouse is organized around major subjects of the enterprise rather than around applications.

- Focuses on subjects like **sales, customer, product**, etc.
- Supports analysis and decision making.

Example:
Instead of organizing data by operational processes, it may organize data around the subject **"sales"** for analysis.

---

2) Integrated

Data in a warehouse is collected from multiple heterogeneous sources and stored in a unified format.

- Different sources may use different naming conventions, formats, and codes.
- These inconsistencies are resolved before storing in the warehouse.

Example:
If two systems identify a product differently, the warehouse standardizes it into a single consistent format.

---

3) Time-Variant

Data warehouse stores historical data over a long period.

- Data is maintained with time references.
- Used for trend analysis and forecasting.

Example:
Users can retrieve sales data from:

- 3 months ago
- 6 months ago
- 1 year ago

Unlike OLTP systems, which mostly store current data.

---

4) Non-Volatile

Once data is stored in the warehouse, it is not changed or deleted.

- Data is read-only.
- No frequent updates like in OLTP systems.
- Only loading and accessing are performed.

This ensures consistency for decision-making.

Final Answer (Exam Format – 5 Marks)

The four characteristics of a Data Warehouse are:

1. **Subject-Oriented** – Organized around major business subjects such as sales or customers.
2. **Integrated** – Data from multiple sources is cleaned and unified into a consistent format.
3. **Time-Variant** – Stores historical data over long periods for analysis.
4. **Non-Volatile** – Data once entered is not updated or deleted; it is stable and read-only.

3.Difference b/w OLAP and OLTP.

Comparison of OLTP and OLAP:

Features	**OLTP**	**OLAP**
Characteristic	operational processing	informational processing
Orientation	Transaction	Analysis
User	Clients, clerks, IT professionals	knowledge worker (e.g., manager executive, analyst)
DB design	entity-relationship (ER) model	star/snowflake
Data	Current data	historic
Summarization	primitive, highly detailed	summarized, consolidated
Unit of work	short, simple transaction	complex query
Access	read/write	mostly read
Focus	data in	information out

Number of records Accessed	Tens	Millions
Number of users	Thousands	Hundreds
DB size	GB to high-order GB	>=TB

4.ODS(Operational Data Store)

What is ODS?

An **Operational Data Store (ODS)** is a database that:

- Integrates data from multiple operational systems
- Stores **current, near real-time data**
- Supports operational reporting (not long-term strategic analysis)

It acts as an **intermediate layer** between OLTP systems and the Data Warehouse.

Why is ODS needed?

In data warehousing architecture (Module 1 – Multitier Architecture section), data is extracted from operational databases and processed before loading into the warehouse.

An ODS is often used:

Operational Systems → ODS → Data Warehouse

It helps in:

- Data cleaning
- Data integration
- Providing up-to-date operational reports

Difference Between ODS and Data Warehouse

Feature	ODS	Data Warehouse

Data Type	Current data	Historical data
Update	Frequently updated	Non-volatile (no updates)
Purpose	Operational reporting	Strategic decision making
Data Detail	Detailed	Summarized + detailed
Time Period	Short-term	Long-term

Simple Understanding

Think of it like this:

- **OLTP** → Handles daily transactions
- **ODS** → Stores cleaned current data for quick operational reporting
- **Data Warehouse** → Stores historical data for analysis and decision making

---

Important Point for Exams

If asked in exam:

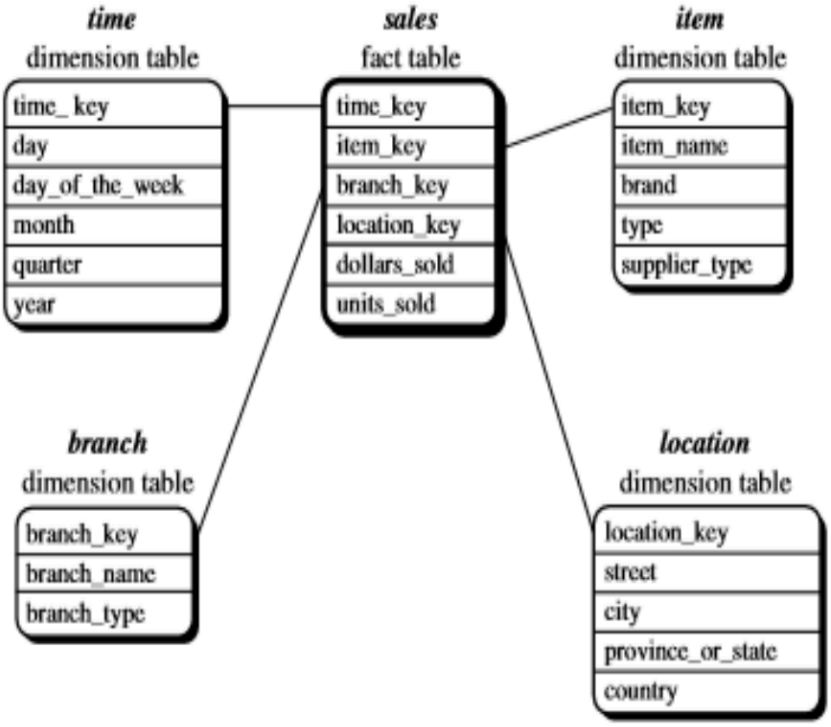
> ODS is used for operational reporting and contains current integrated data, while Data Warehouse is used for analytical processing and contains historical non-volatile data.

5.Snowflake and Star Schema

i. Star schema.

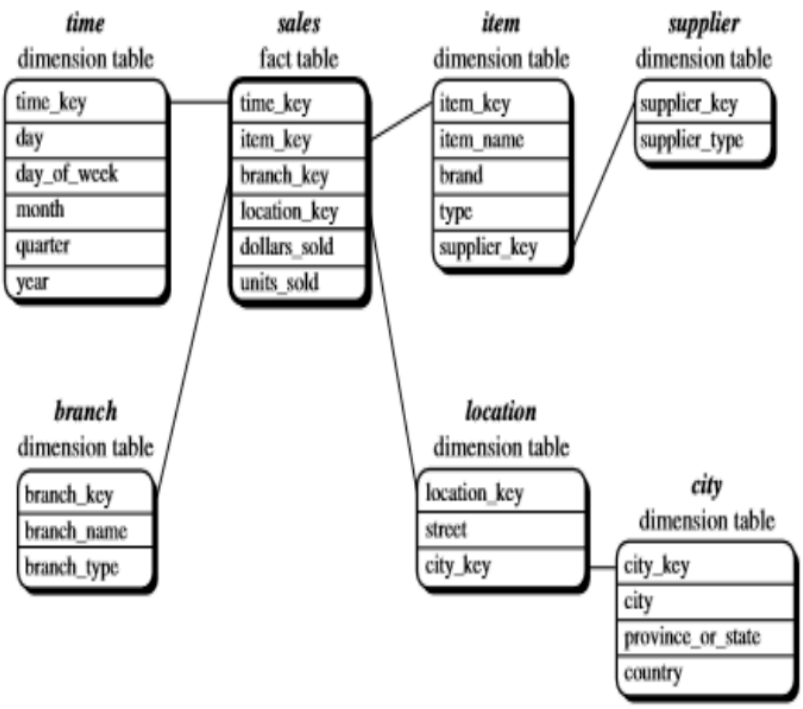
A star schema for All Electronics sales is shown in Figure

Sales are considered along four dimensions: time, item, branch, and location. The schema contains a central fact table for sales that contains keys to each of the four dimensions, along with two measures: dollars sold and units sold. To minimize the size of the fact table, dimension identifiers (e.g., time key and item key) are system-generated identifiers.



ii. Snowflake schema:

- The snowflake schema is a variant of the star schema model, where some dimension tables are normalized, thereby further splitting the data into additional tables.
- The resulting schema graph forms a shape similar to a snowflake.
- The major difference between the snowflake and star schema models is that the dimension tables of the snowflake model may be kept in normalized form to reduce redundancies.
- Such a table is easy to maintain and saves storage space. However, this space savings is negligible in comparison to the typical magnitude of the fact table.



Difference Between Star and Snowflake Schema

Feature	Star Schema	Snowflake Schema
Structure	Fact + Dimension tables	Fact + Sub-dimension tables
Normalizatio n	No normalization	Uses normalization
Design	Simple	Complex
Space Usage	More	Less
Foreign Keys	Fewer	More

6.Data Mining

Definition

Data Mining is the process of discovering interesting patterns and knowledge from large amounts of data.

It is also referred to as part of **Knowledge Discovery in Databases (KDD)** — which is the overall process of converting raw data into useful information.

Data Mining as Part of KDD

According to Module 2, Data Mining is **one step in the KDD process**, which includes:

1. Data Cleaning
2. Data Integration
3. Data Selection
4. Data Transformation
5. **Data Mining** (pattern extraction step)
6. Pattern Evaluation
7. Knowledge Presentation

So, Data Mining is not the entire process — it is the core analytical step inside KDD.

Major Data Mining Tasks

Data mining tasks are divided into two categories:

1. Predictive Tasks

Used to predict unknown or future values.

- **Classification** → For discrete values
  Example: Predict whether a customer will buy a product.
- **Regression** → For continuous values
  Example: Predict stock price.

---

2. Descriptive Tasks

Used to find patterns that describe the data.

Core descriptive tasks:

- **Association Analysis**
  Example: {Diapers} → {Milk}
- **Cluster Analysis**
  Grouping similar objects together.
- **Anomaly Detection**
  Detect unusual or rare patterns (e.g., fraud detection).

Motivating Challenges of Data Mining

From Module 2, major challenges include:

- Scalability (handling huge datasets)

- High dimensionality
- Heterogeneous and complex data
- Distributed data
- Need for automated pattern discovery

Applications of Data Mining

- Market basket analysis
- Fraud detection
- Credit card risk analysis
- Customer segmentation
- Medical diagnosis

Short Exam Answer (5 Marks)

Data Mining is the process of extracting useful patterns and knowledge from large datasets. It is an essential step in the KDD process. Data mining tasks are broadly classified into predictive tasks (classification and regression) and descriptive tasks (association, clustering, anomaly detection).

7.Task and Motivating Challenges of Data Mining

1. Data Mining Tasks

Data mining tasks are broadly classified into **two categories**:

A) Predictive Tasks

Objective: Predict the value of a target (dependent) variable using other attributes.

Types:

• Classification

- Used for **discrete values**
- Example: Predict whether a customer will purchase (Yes/No)

• Regression

- Used for **continuous values**
- Example: Predict income or stock price

B) Descriptive Tasks

Objective: Discover patterns that describe relationships in data.

Core Descriptive Tasks:

• Association Analysis

Finds interesting relationships between items.
Example: {Diapers} → {Milk}

• Cluster Analysis

Groups similar data objects together.

• Anomaly Detection

Identifies unusual or rare patterns.
Example: Fraud detection

---

2. Motivating Challenges of Data Mining

Data mining faces several practical challenges:

1) Scalability

- Modern datasets are very large (GBs, TBs).
- Algorithms must handle massive data efficiently.

---

2) High Dimensionality

- Data may contain hundreds or thousands of attributes.
- Computational complexity increases as dimensions increase.
- Known as the "curse of dimensionality".

---

3) Heterogeneous and Complex Data

- Data may include text, images, videos, graphs, etc.
- Traditional methods handle only structured data.

---

4) Data Ownership and Distribution

- Data is often geographically distributed.
- Owned by different organizations.
- Requires distributed data mining techniques.

---

5) Non-Traditional Analysis

- Traditional approach is hypothesis testing.
- Data mining automates hypothesis generation and pattern discovery.

Exam-Ready Answer (6–8 Marks)

Data mining tasks are classified into predictive tasks (classification and regression) and descriptive tasks (association analysis, clustering, anomaly detection).

The motivating challenges of data mining include scalability, high dimensionality, heterogeneous data, distributed data ownership, and the need for automated pattern discovery.

8.KDD (Knowledge Discovery in Databases)

Definition

KDD (Knowledge Discovery in Databases) is the overall process of converting raw data into useful and meaningful knowledge.

Data Mining is **one important step** in the KDD process — but KDD includes several steps before and after data mining.

KDD Process Steps

According to Module 2, the KDD process consists of the following steps:

1Data Cleaning

- Removes noise and inconsistent data.

2Data Integration

- Combines data from multiple sources.

3Data Selection

- Retrieves relevant data for analysis.

4Data Transformation

- Converts data into suitable format for mining.
- May include aggregation and summarization.

5Data Mining

- Core step where intelligent techniques are applied
- Extracts patterns from data.

6Pattern Evaluation

- Identifies truly interesting and useful patterns.

7Knowledge Presentation

- Presents discovered knowledge using visualization or reports.

Important Point

KDD is an **iterative process**, meaning steps can be repeated for better results.

Difference Between KDD and Data Mining

KDD	Data Mining
Complete process	One step in KDD

Includes preprocessing and postprocessing	Focuses only on pattern extraction
Converts raw data into knowledge	Extracts patterns

Exam-Ready Answer (5–6 Marks)

KDD (Knowledge Discovery in Databases) is the overall process of extracting useful knowledge from large datasets. It includes steps such as data cleaning, integration, selection, transformation, data mining, pattern evaluation, and knowledge presentation. Data mining is a core step within the KDD process.

9.Types of Attributes

Different types of attributes:

- Nominal
- Ordinal
- Interval
- Ratio

Nominal Attributes

- Nominal means "relating to names."
- The values of a nominal attribute are just names of things
- Nominal values provide only enough information to distinguish one object from another(=, ≠)

Example:

- Suppose that hair color and marital status are two attributes
- In our application, possible values for hair color are black, brown, blond, red, grey, and white.
- The attribute marital status can take on the values single, married, divorced, and widowed. Both hair color and marital status are nominal attributes.
- Another example of a nominal attribute is occupation, with the values teacher, dentist, programmer, farmer, and so on.

Ordinal Attributes (<, >, >=, <=)

- An ordinal attribute is an attribute with possible values that have a meaningful order or ranking among them

- ☐ Suppose that drink size corresponds to the size of drinks available at a fast-food restaurant. This attribute has three possible values: small, medium, and large. The values have a meaningful sequence
- ☐ Other examples of ordinal attributes include grade (e.g., A+, A, A−, B+, and so on)
- ☐ Professional ranks can be enumerated in a sequential order: for example, assistant, associate, specialist
- ☐ Customer satisfaction has the following ordinal categories: 0: very dissatisfied, 1: somewhat dissatisfied, 2: neutral, 3: satisfied, and 4: very satisfied.

Interval-Scaled Attributes (+ or -)

- ☐ Interval-scaled attributes are measured on a scale of equal-size units
- ☐ Attributes allow us to compare and quantify the difference between values.
- ☐ For example, a temperature of 20○C is five degrees higher than a temperature of 15○C.
- ☐ Calendar dates are another example. For instance, the years 2002 and 2010 are eight years apart.

Ratio scaled attributes (* or /)

- ☐ A measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value
- ☐ Years of experience (e.g., the objects are employees) and number of words (e.g., the objects are documents).